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Z': A Proposed Directed Output of the TADA Model

SNP-to-gene pipeline via MAGMA in a common variant GWAS leads to principled pathway analysis.

Multi-marker Analysis of GenoMic Annotation (MAGMA) is a gene-level and gene-set analysis tool where at its core is a multiple regression model defined by

$$Y = \alpha_{0,g} + X_g\alpha_g + W\beta_g + \epsilon_g$$

where Y is a vector of summary Z statistics per SNP, dependent upon: X_g, a matrix of SNP genotypes, α_g, the effect of interest, W, a matrix of covariates, β_g, the coefficients of the covariates, and ε_g, the residual error, striated across each gene g. The test for gene-level association assesses whether the joint effect of SNPs across the gene is significantly nonzero, using an F-test or equivalent. The SNP-wise mean is calculated in which each gene-level p-value is converted to a Z-score via

$$Z_g = \hat{\alpha}_g \cdot \Phi^{-1}(1 - p_g)$$

where Φ^{-1} is the inverse normal (probit) transformation and p_g is the gene-level p-value from the regression. This Z-score can thus be modeled by the full frequentist definition of a Z-statistic:

$$Z = \frac{\hat{\theta} - \theta_0}{\mathrm{SD}(\hat{\theta})}$$

The Z-statistic is preferable given its directionality, as dictated by the parameter magnitude in reference to its value under the null. Moreover, the sign of α_g preserves the SNP effects in the calculation of the Z score, preserving the biological relevance of gene enrichment (+ Z) and depletion (-Z). Therefore, using this statistic is ideal for downstream pathway analysis to rank genes of interest by both magnitude and direction. This model, however, works mainly in conjunction with common-variant GWAS data, as rare-variant analyses typically require other statistical methods to compensate for lower power under the same assumptions.

Managing statistical power in rare-variant analysis leaves biological relevance in pathway analysis behind.

In the instance of ASD research, the probabilistic Transmission And De novo Association (TADA) model has been developed using a Bayesian statistical framework to prioritize risk genes from WES due to *de novo* mutations, inherited variants, and from case-control data.

Specifically, it is a gene-based likelihood model using data-agnostic parameters that would otherwise be difficult to extract per gene. In adaptations such as TADA+, '+' refers to the incorporation of functional genomic annotations as informative priors, allowing biologically relevant features to up- or down-weight genes by the potential functional severity of their SNPs (e.g. loss-of-function > missense > synonymous). In either instance, the resulting q-value is a directionless metric bounded by 0 and 1, and the number of unassociated genes skews the q-value distribution toward 1. Relatively data-driven q-value thresholds have been identified to categorize regions of the post-cutoff distribution by strength of association with ASD. For example, the remaining data from a q-threshold of 0.1 represents the retention of ASD-suggestive genes, used in exploratory analysis.While innovative, the output of this model is not ideal for downstream pathway analysis as the MAGMA Z-statistic given the lack of depletion representation, removing effect as information of interest.

Here we seek to take advantage of the Bayesian marginal likelihood composition of the TADA model to extract a Z-like statistic, Z'. We hypothesize that its use as a ranking method in downstream pathway enrichment analysis will maintain biological relevance and corroborate that both gene enrichment and depletion in ASD rare-variant analysis is statistically tractable. Initially, we will calculate Z' using *de novo* variants as a proof-of-concept before expanding to other mutation types.

Deriving Z'.

Given the data are discrete rare-variant counts whose exposure measure is their mutation rate over time, each data point is distributed as a Poisson random variable X_i:

$$X_i \sim \text{Poisson}\left(2n \cdot \mu_i \cdot \gamma^{I(\text{case})}\right)$$

where the factor of 2 refers to chromosome pairs, n is the number of trios (patient-parent samples), μ_i is the mutation rate per gene, and γ is the relative risk parameter determined by the indicator function

$$I(\text{case}) = \begin{cases} 1 & \text{if case} \\ 0 & \text{if control} \end{cases}$$

Thus we define γ as a multiplicative factor that scales the baseline in cases relative to controls. Under the null, $\gamma = 1$, with no meaningful differentiation in risk between cases and controls.

The full likelihood is structured based upon the mixed likelihoods of relative risk: $P(x_i \mid H_1)$ and $P(x_i \mid H_0)$ weighted by the proportion of genes in the genome affecting ASD risk, π versus 1 - π, respectively:

$$P(x_i) = \pi \, P(x_i \mid H_1) + (1 - \pi) \, P(x_i \mid H_0)$$

The null marginal is a Poisson likelihood conditioning on the null relative risk value, $\gamma = 1$,

$$P(x_i \mid H_0) = p(x_i \mid \gamma_i = 1) = \mathrm{Poisson}(x_i \mid c_i), \quad c_i = 2n\mu_i$$

while the alternative marginal likelihood is the integrated product of the Poisson alternative likelihood and a gamma prior placed on the relative risk parameter:

$$P(x_i \mid H_1) = \int p(x_i \mid \gamma_i) \cdot p(\gamma_i \mid H_1) \, d\gamma_i$$

The gamma prior can be written as

$$\gamma_i \sim \mathrm{Gamma}(\bar{\gamma}\beta, \ \beta) \implies f(\gamma) = \frac{\beta^{\bar{\gamma}\beta}}{\Gamma(\bar{\gamma}\beta)} \cdot \gamma^{\bar{\gamma}\beta - 1} \cdot e^{-\gamma\beta}$$

with $\bar{\gamma}$ and β as our smoothed mean relative risk and rate hyperparameters, respectively, and the alternative marginal likelihood is fully shown as

$$p(x_i \mid \gamma_i) = \frac{(c_i\gamma_i)^{x_i} \, e^{-c_i\gamma_i}}{x_i!}$$

We recognize that we can derive the alternative marginal likelihood from the Gamma-independent kernel and the resulting alternative conditional posterior by marginalizing out γ. To start, we show the complete joint expression under the integral and its simplification

$$f(\gamma) \cdot p(x_i \mid \gamma_i) = \frac{\beta^{\bar{\gamma}\beta}}{\Gamma(\bar{\gamma}\beta)} \cdot \gamma_i^{\bar{\gamma}\beta - 1} \cdot e^{-\gamma_i \beta} \cdot \frac{(c_i \gamma_i)^{x_i} \, e^{-c_i \gamma_i}}{x_i!}$$

$$= \frac{\beta^{\bar{\gamma}\beta} \, c_i^{x_i}}{x_i! \, \Gamma(\bar{\gamma}\beta)} \cdot \gamma_i^{(x_i + \bar{\gamma}\beta - 1)} \cdot e^{-\gamma_i(\beta + c_i)}$$

deriving our new shape and and rate parameters for the alternative conditional posterior via the Gamma-Poisson conjugacy

$$\gamma_i \mid x_i, H_1 \sim \text{Gamma}(\bar{\gamma}\beta + x_i, \;\; \beta + c_i)$$

In parallel, we integrate the Gamma-independent kernel over all values of x_i to get a Negative Binomial PMF with mapped shape and rate parameters, yielding the alternative marginal likelihood

$$P(x_i \mid H_1) = \text{NegBinom}\left(x_i \,\middle|\, \bar{\gamma}\beta, \; \frac{\beta}{\beta + c_i}\right)$$

From this point, we achieve the full mixed posterior

$$p(\gamma_i \mid x_i) = (1 - w_i)\,\delta(\gamma_i - 1) + w_i \cdot \text{Gamma}\left(\gamma_i \,|\, \bar{\gamma}\beta + x_i, \; \beta + c_i\right)$$

where the Dirac delta references to the distributionless null value set at 1, and the weights w_i calculated from Bayes' Rule and the Law of Total Probability:

$$w_i = \frac{\pi\, P(x_i \mid H_1)}{\pi\, P(x_i \mid H_1) + (1-\pi)\, P(x_i \mid H_0)}$$

more specifically written as constructed from our previously-derived likelihoods

$$w_i = \frac{\pi \cdot \mathrm{NegBinom}}{\pi \cdot \mathrm{NegBinom} + (1-\pi) \cdot \mathrm{Poisson}}$$

Finally, we can achieve our Z’ by finding the weighted posterior mean (expectation)

$$\mathbb{E}[\gamma_i \mid x_i] = (1 - w_i)(1) + w_i \left(\frac{\bar{\gamma}\beta + x_i}{\beta + c_i} \right)$$

and variance

$$\mathrm{Var}(\gamma_i \mid x_i) = w_i \left(\frac{\bar{\gamma}\beta + x_i}{(\beta + c_i)^2} \right) + w_i(1 - w_i) \left[\left(\frac{\bar{\gamma}\beta + x_i}{\beta + c_i} \right) - 1 \right]^2$$

from the mixed distribution, centered around the null posterior mean 1.

In simplified terms, Z’ can be written as

$$Z_i' = \frac{\mathbb{E}[\gamma_i \mid x_i] - 1}{\sqrt{\mathrm{Var}(\gamma_i \mid x_i)}}$$

Upon derivation, it was found that remaining on the natural scale rather than log transforming was optimal as $\gamma = 1$ is a satisfying biological null, and a log transform would obscure the very directionality we seek to retain. Thus, our Z' result has the following definition of directionality:

$$Z' \begin{cases} > 0 & \hat{\gamma} > 1 \quad \text{enrichment} \\ = 0 & \hat{\gamma} = 1 \quad \text{null} \\ < 0 & \hat{\gamma} < 1 \quad \text{depletion} \end{cases}$$

Proposal.

The intended purpose of this Z'-statistic is for it to contextualize gene-based ASD risk likelihood with a biologically relevant output. Specifically, we seek to rank these genes by their Z' and feed them into FGSEA, effectively replacing the q-value. Validation of the statistic will compare the enrichment results from genes that are either q- or Z'-ranked, determining if (1) both enrichment and depletion are represented and (2) the distribution is not as drastically right-skewed. Ultimately, the generation of this Z'-statistic could further propagate ASD research in the cross-variant pathway analysis space and decrease the chance of erroneous assumptions about underlying ASD biology. The script for calculation and validation of this statistic can be found in the asmurrill29/asd-risk-stat Github repository.

Assumptions and limitations.

Although this calculation is closed form and requires no mathematical approximations, some assumptions must hold for the statistic to be robust. As follows from previous ASD variant-class research (asmurrill29/common-vs-rare-variants-ASD), a comparison between a MAGMA Wald-Z output with a TADA Z' output for pathway analysis across variant class would not map one-to-one, given that our Z-like statistic does not follow a perfectly normal distribution. However, as the second variance term $w_i(1 - w_i)(\mu_{H1} - 1)^2$ shrinks Z' for genes of ambiguous risk, it conversely produces large magnitudes for extreme cases of risk. This gives evidence for the statistic behaving sensibly even without a guarantee of normality.

CITATIONS

1. He, X., Sanders, et al. (2013). Integrated Model of De Novo and Inherited Genetic Variants Yields Greater Power to Identify Risk Genes. PLoS Genetics, 9(8), e1003671. https://doi.org/10.1371/journal.pgen.1003671
2. Satterstrom, F. K. *et al*. (2020). Large-Scale Exome Sequencing Study Implicates Both Developmental and Functional Changes in the Neurobiology of Autism. Cell, 180(3) 568–584.